

T20 DataHub

API Documentation

March 2026 · Moksha Premesh Kancharla · COMP3011 Coursework

Swagger UI	https://t20-datahub-api.onrender.com/docs
Frontend	https://t20-datahub-api.onrender.com
Auth	JWT Bearer Token (Authorization: Bearer)
Data Format	JSON (Content-Type: application/json)
Dataset	216 T20 World Cup matches (2014–2026) · Cricsheet.org (CC BY 4.0)
Endpoints	23 endpoints across 6 resource groups

Table of Contents

1. Overview & Authentication
2. Auth Endpoints (/auth/register · /auth/login · /auth/me · /auth/make-admin)
3. Match Endpoints (/matches · /matches/{id} · /matches/{id}/scorecard · /squads)
4. Admin — Match Management (POST · PATCH · DELETE /admin/matches)
5. Fan Profile & Dashboard (/me/profile · /me/dashboard)
6. Fantasy Cricket (/fantasy/teams · captain · vice-captain · leaderboards)
7. Reference Data & Health Check (/options/* · /health)
8. Error Codes Reference

1. Overview & Authentication

The T20 DataHub API is a RESTful service built with FastAPI and PostgreSQL, providing ball-by-ball T20 World Cup cricket data from 2014 to 2026. All responses are JSON. A React 18 SPA frontend is served directly by FastAPI in production from frontend/dist/.

Protected endpoints require a valid JWT Bearer token in the Authorization header, obtained via POST /auth/login. Tokens expire after 24 hours. Interactive documentation and testing is available at the Swagger UI (/docs).

Authentication Flow

```
# Step 1: Register
POST /auth/register
Content-Type: application/json
{"username": "fan1", "email": "fan@example.com", "password": "cricket123"}

# Step 2: Login (form-encoded, NOT JSON)
POST /auth/login
Content-Type: application/x-www-form-urlencoded
username=fan1&password=cricket123
# → {"access_token": "eyJ...", "token_type": "bearer"}

# Step 3: Use token on protected endpoints
GET /me/profile
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

Password Policy

Rule	Requirement
Minimum length	10 characters
Must contain	At least one letter and one number
Rejected passwords	Common/weak passwords rejected (e.g. password123)
Token lifetime	24 hours
Rate limiting	Login endpoint: max 10 requests/minute per IP

Admin Access

To gain admin access, call POST /auth/make-admin?secret=<ADMIN_SECRET> while authenticated. The secret is configured via the ADMIN_SECRET environment variable on the server.

2. Auth Endpoints

POST /auth/register Register a new user

Creates a new user account. Username and email must be unique. Password must satisfy the policy above.

Request Body

Parameter	Type	Required	Description
username	string	Yes	Unique username
email	string	Yes	Valid email address
password	string	Yes	Min 10 chars, must contain letter + number

Example Request

```
POST /auth/register
Content-Type: application/json
{
  "username": "cricket_fan_99",
  "email": "fan@example.com",
  "password": "cricket123"
}
```

Example Response — 201 Created

```
{
  "id": 42,
  "username": "cricket_fan_99",
  "email": "fan@example.com",
  "role": "user",
  "created_at": "2026-03-11T10:00:00Z"
}
```

Response Codes

Code	Status	Description
201	Created	User successfully registered
400	Bad Request	Weak password or password policy violation
409	Conflict	Username or email already exists
422	Unprocessable Entity	Invalid email format or missing fields

POST /auth/login Login and receive JWT token

Authenticates a user and returns a JWT access token. The request body must be form-encoded (application/x-www-form-urlencoded), not JSON. Rate limited to 10 requests/minute per IP.

Request Body (form-encoded)

Parameter	Type	Required	Description
username	string	Yes	Registered username
password	string	Yes	Account password

Example Request

```
POST /auth/login
Content-Type: application/x-www-form-urlencoded
username=cricket_fan_99&password=cricket123
```

Example Response — 200 OK

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer"
}
```

Response Codes

Code	Status	Description
200	OK	Login successful, token returned
401	Unauthorised	Incorrect username or password
429	Too Many Requests	Rate limit exceeded (10 req/min per IP)

GET /auth/me Get current authenticated user

Requires authentication (Bearer token)

Returns the authenticated user profile information.

Example Response — 200 OK

```
{
  "id": 42,
  "username": "cricket_fan_99",
  "email": "fan@example.com",
  "role": "user"
}
```

Response Codes

Code	Status	Description
200	OK	User data returned
401	Unauthorised	Missing or invalid token

POST /auth/make-admin Promote yourself to admin

Requires authentication (Bearer token)

Promotes the currently authenticated user to the admin role. Requires the ADMIN_SECRET query parameter matching the server-side environment variable.

Query Parameters

Parameter	Type	Required	Description
secret	string	Yes	Must match ADMIN_SECRET environment variable

Example Request

```
POST /auth/make-admin?secret=your-admin-secret
Authorization: Bearer <token>
```

Response Codes

Code	Status	Description
200	OK	User promoted to admin
401	Unauthorised	Not authenticated
403	Forbidden	Invalid admin secret

3. Match Endpoints

GET /matches List all matches with optional filters

Returns a paginated list of T20 World Cup matches. Supports filtering by year, team, stage, and venue. Contains 216 matches from 2014, 2016, 2021, 2022, 2024, and 2026 tournaments. No authentication required.

Query Parameters

Parameter	Type	Required	Description
year	integer	No	Filter by tournament year (e.g. 2024)
team	string	No	Filter by team name — matches team1 or team2
stage	string	No	Filter by stage: Final, Semi Final, Super 10, First Round
venue	string	No	Partial venue name match
limit	integer	No	Max results to return (default: 100)
offset	integer	No	Pagination offset (default: 0)

Example Request

```
GET /matches?year=2024&team=India&limit=10
```

Example Response — 200 OK

```
{
  "total": 8, "returned": 8,
  "filters": {"team": "India", "year": "2024", "stage": null, "venue": null},
  "matches": [{
    "id": 123, "match_date": "2024-06-29",
    "team1": "India", "team2": "South Africa",
    "venue": "Kensington Oval, Bridgetown, Barbados",
    "stage": "Final", "tournament_year": "2024", "winner": "India",
    "innings_scores": [
      {"team": "India", "runs": 176, "wickets": 7, "overs": "20.0"},
      {"team": "South Africa", "runs": 169, "wickets": 8, "overs": "20.0"}
    ]
  }]
}
```

Response Codes

Code	Status	Description
200	OK	Match list returned
422	Unprocessable Entity	Invalid query parameter type

GET `/matches/{id}` Single match detail

Returns summary details for a single match by ID.

Path Parameters

Parameter	Type	Required	Description
id	integer	Yes	The unique match ID

Response Codes

Code	Status	Description
200	OK	Match detail returned
404	Not Found	Match ID does not exist

GET `/matches/{id}/scorecard` Full batting and bowling scorecard

Returns a detailed scorecard including batting, bowling, and fall of wickets for each innings.

Path Parameters

Parameter	Type	Required	Description
id	integer	Yes	The unique match ID

Example Request

```
GET /matches/123/scorecard
```

Example Response — 200 OK

```
{
  "match_id": 123,
  "team1": "India", "team2": "South Africa",
  "match_date": "2024-06-29", "winner": "India",
  "innings": [{
    "team": "India", "total_runs": 176, "total_wickets": 7,
    "batting": [{"batter": "Rohit Sharma", "runs": 9, "balls": 11,
      "fours": 1, "sixes": 0, "strike_rate": "81.82",
      "dismissal": "c Miller b Ngidi"}],
    "bowling": [{"bowler": "Jasprit Bumrah", "overs": "4.0",
      "maidens": 0, "runs": 18, "wickets": 2, "economy": "4.50"}]
  }]
}
```

Response Codes

Code	Status	Description
200	OK	Scorecard returned
404	Not Found	Match ID does not exist

GET `/matches/{id}/squads` Match squads for player selection

Returns available players for both teams. Always call this before building a fantasy team — player names submitted to POST /fantasy/teams must exactly match names returned here.

Path Parameters

Parameter	Type	Required	Description
id	integer	Yes	The unique match ID

Example Response — 200 OK

```
{
  "match_id": 123,
  "India": [
    {"name": "Rohit Sharma", "role": "Batsman"},
    {"name": "Jasprit Bumrah", "role": "Bowler"}
  ],
  "South Africa": [
    {"name": "Quinton de Kock", "role": "WK-Batsman"},
    {"name": "Kagiso Rabada", "role": "Bowler"}
  ]
}
```

Response Codes

Code	Status	Description
200	OK	Squad data returned
404	Not Found	Match not found

4. Admin — Match Management

Admin endpoints allow creating, updating, and deleting match records. All require an authenticated user with the admin role. Non-admin users receive 403 Forbidden. Admin routes are prefixed `/admin/matches`.

POST `/admin/matches` Create a new match (Admin)

Admin role required

Request Body

Parameter	Type	Required	Description
team1	string	Yes	First team name
team2	string	Yes	Second team name
match_date	string	Yes	Date in YYYY-MM-DD format
venue	string	Yes	Venue name
tournament_year	string	Yes	Tournament year (e.g. '2026')
winner	string	No	Winning team (null for no result)
toss_winner	string	No	Team that won the toss
stage	string	No	Match stage (Final, Semi Final, etc.)

Example Request

```
POST /admin/matches
Authorization: Bearer <admin_token>
Content-Type: application/json
{
  "team1": "India", "team2": "Australia",
  "match_date": "2026-03-15",
  "venue": "Eden Gardens, Kolkata",
  "tournament_year": "2026",
  "winner": "India", "stage": "Final"
}
```

Response Codes

Code	Status	Description
201	Created	Match successfully created
401	Unauthorised	Missing or invalid token
403	Forbidden	User is not an admin
422	Unprocessable Entity	Missing required fields or invalid data

PATCH `/admin/matches/{id}` Update match fields (Admin)

Admin role required

Partially updates a match record. Only fields provided in the request body are updated; omitted fields remain unchanged.

Path Parameters

Parameter	Type	Required	Description
id	integer	Yes	ID of the match to update

Example Request

```
PATCH /admin/matches/123
Authorization: Bearer <admin_token>
Content-Type: application/json
{"winner": "South Africa"}
```

Response Codes

Code	Status	Description
200	OK	Match updated, returns updated match object
401	Unauthorised	Missing or invalid token
403	Forbidden	Not an admin
404	Not Found	Match ID does not exist

DELETE `/admin/matches/{id}` Delete a match (Admin)

Admin role required

Path Parameters

Parameter	Type	Required	Description
id	integer	Yes	ID of the match to delete

Response Codes

Code	Status	Description
204	No Content	Match deleted successfully
401	Unauthorised	Missing or invalid token
403	Forbidden	Not an admin
404	Not Found	Match ID does not exist

5. Fan Profile & Dashboard

Fan profile endpoints allow authenticated users to create and manage a personal cricket profile with favourite team, player, and a bio. Each user may have at most one profile. The dashboard returns personalised charts and stats.

PUT `/me/profile` Create or update fan profile

Requires authentication (Bearer token)

Creates a profile if one does not exist, or updates it if it does. All fields are optional.

Request Body

Parameter	Type	Required	Description
favourite_team	string	No	Favourite T20 team name
favourite_player	string	No	Favourite player name
bio	string	No	Short personal bio

Example Response — 200 OK

```
{
  "id": 1, "user_id": 42,
  "favourite_team": "India",
  "favourite_player": "Virat Kohli",
  "bio": "Cricket fan since 2007"
}
```

Response Codes

Code	Status	Description
200	OK	Profile created or updated
401	Unauthorised	Not authenticated

GET /me/profile Get fan profile

Requires authentication (Bearer token)

Response Codes

Code	Status	Description
200	OK	Profile returned
401	Unauthorised	Not authenticated
404	Not Found	No profile exists for this user

DELETE /me/profile Delete fan profile

Requires authentication (Bearer token)

Response Codes

Code	Status	Description
204	No Content	Profile deleted
401	Unauthorised	Not authenticated
404	Not Found	No profile exists

GET /me/dashboard Personalised dashboard

Requires authentication (Bearer token)

Returns personalised stats including favourite team win/loss charts, player career history, fantasy team count, and total matches in the database.

Example Response — 200 OK

```
{
  "username": "cricket_fan_99",
  "favourite_team": "India",
  "fantasy_teams": 2,
  "total_matches_in_db": 216
}
```

Response Codes

Code	Status	Description
200	OK	Dashboard data returned
401	Unauthorised	Not authenticated

6. Fantasy Cricket

Users build an XI from real match squads, assign a captain (2x points) and vice-captain (1.5x points), submit the team for scoring, and compete on per-match and global leaderboards. Player names must exactly match strings returned by GET /matches/{id}/squads.

GET /matches/{id}/fantasy/teams List my fantasy teams for a match

Requires authentication (Bearer token)

Path Parameters

Parameter	Type	Required	Description
id	integer	Yes	Match ID

Example Response — 200 OK

```
[
  {
    "id": 1, "name": "My Dream XI",
    "captain": "Virat Kohli", "vice_captain": "Rohit Sharma",
    "player_count": 11, "total_points": 245.5
  }
]
```

Response Codes

Code	Status	Description
200	OK	Team list returned
401	Unauthorised	Not authenticated

POST /matches/{id}/fantasy/teams Create a new fantasy team

Requires authentication (Bearer token)

Request Body

Parameter	Type	Required	Description
name	string	Yes	Team name
players	array[string]	Yes	Exactly 11 player names from the match squads
captain	string	Yes	Captain (must be in players list) — 2x points
vice_captain	string	Yes	Vice-captain (must differ from captain) — 1.5x points

Example Request

```
POST /matches/123/fantasy/teams
Authorization: Bearer <token>
Content-Type: application/json
{
  "name": "My Dream XI",
  "players": ["Rohit Sharma", "Virat Kohli", "...9 more..."],
  "captain": "Virat Kohli",
  "vice_captain": "Rohit Sharma"
}
```

Response Codes

Code	Status	Description
201	Created	Fantasy team created
400	Bad Request	Invalid players, wrong count, or captain not in squad
401	Unauthorised	Not authenticated

GET `/fantasy/teams/{id}` Get a specific fantasy team

Requires authentication (Bearer token)

Path Parameters

Parameter	Type	Required	Description
id	integer	Yes	Fantasy team ID

Response Codes

Code	Status	Description
200	OK	Team details returned
401	Unauthorised	Not authenticated
403	Forbidden	Not your team
404	Not Found	Team not found

PUT `/fantasy/teams/{id}/captain` Set captain (2x points)

Requires authentication (Bearer token)

Path Parameters

Parameter	Type	Required	Description
id	integer	Yes	Fantasy team ID

Response Codes

Code	Status	Description
200	OK	Captain updated
401	Unauthorised	Not authenticated
403	Forbidden	Not your team
404	Not Found	Team not found

PUT `/fantasy/teams/{id}/vice-captain` Set vice-captain (1.5x points)

Requires authentication (Bearer token)

Path Parameters

Parameter	Type	Required	Description
id	integer	Yes	Fantasy team ID

Response Codes

Code	Status	Description
200	OK	Vice-captain updated
401	Unauthorised	Not authenticated
403	Forbidden	Not your team
404	Not Found	Team not found

POST `/fantasy/teams/{id}/submit` Submit team for scoring

Requires authentication (Bearer token)

Locks the fantasy team and calculates points based on real match performance. Captain scores 2x, vice-captain 1.5x.

Path Parameters

Parameter	Type	Required	Description
id	integer	Yes	Fantasy team ID

Response Codes

Code	Status	Description
200	OK	Team submitted, points calculated
400	Bad Request	Team already submitted or invalid
401	Unauthorised	Not authenticated
403	Forbidden	Not your team

DELETE `/fantasy/teams/{id}` Delete a draft fantasy team

Requires authentication (Bearer token)

Path Parameters

Parameter	Type	Required	Description
id	integer	Yes	Fantasy team ID

Response Codes

Code	Status	Description
204	No Content	Team deleted
401	Unauthorised	Not authenticated
403	Forbidden	Not your team
404	Not Found	Team not found

GET `/matches/{id}/fantasy/leaderboard` Per-match fantasy leaderboard

Returns the fantasy leaderboard for a specific match, ranked by total points.

Example Response — 200 OK

```
[
  {
    "rank": 1,
    "username": "cricket_ace",
    "total_points": 892.5,
  },
  {
    "rank": 2,
    "username": "fan99",
    "total_points": 743.0
  }
]
```

Response Codes

Code	Status	Description
200	OK	Leaderboard returned
404	Not Found	Match not found

GET `/fantasy/leaderboard` Global fantasy leaderboard

Returns the global leaderboard ranked by total fantasy points across all matches and all users.

Example Response — 200 OK

```
[
  {"rank": 1, "username": "cricket_ace", "total_points": 892.5},
  {"rank": 2, "username": "fan99", "total_points": 743.0},
  {"rank": 3, "username": "t20_legend", "total_points": 698.5}
]
```

Response Codes

Code	Status	Description
200	OK	Global leaderboard returned

7. Reference Data & Health Check

GET `/options/years` Tournament years with match data

Returns the list of tournament years that have match data in the database.

Example Response — 200 OK

```
[2014, 2016, 2021, 2022, 2024, 2026]
```

Response Codes

Code	Status	Description
200	OK	Year list returned

GET `/options/teams` All teams with win/loss statistics

Returns all teams in the database with win and loss counts. Used by the frontend to populate filter dropdowns.

Example Response — 200 OK

```
[
  {"team": "India", "wins": 45, "losses": 12},
  {"team": "Australia", "wins": 38, "losses": 18}
]
```

Response Codes

Code	Status	Description
200	OK	Team list with stats returned

GET `/options/players` List all players in the database

Returns all player names across all match records. Used by the frontend for player search and filtering.

Response Codes

Code	Status	Description
200	OK	Player name list returned

GET `/health` API health check

Confirms the API is running. No authentication required. Suitable for uptime monitoring or keep-alive pings on the Render free tier.

Example Response — 200 OK

```
{"status": "ok", "project": "T20DataHub"}
```

Response Codes

Code	Status	Description
200	OK	API is healthy and running

8. Error Codes Reference

All error responses follow a consistent JSON format. The detail field contains a human-readable message. Validation errors (422) return a list of field-level objects with loc, msg, and type fields.

Standard Error Response Format

```
{"detail": "Incorrect username or password"}
```

Validation Error Format (422)

```
{
  "detail": [{
    "loc": ["body", "password"],
    "msg": "Password must be at least 10 characters",
    "type": "value_error"
  }]
}
```

Complete Error Code Reference

Code	Status	When it occurs
200	OK	Successful GET, PUT, PATCH requests
201	Created	Successful POST requests (resource created)
204	No Content	Successful DELETE requests
400	Bad Request	Business logic error (e.g. weak password, team already submitted)
401	Unauthorised	Missing, expired, or invalid JWT token
403	Forbidden	Valid token but insufficient permissions (non-admin, not your resource)
404	Not Found	Resource does not exist (match ID, team ID, profile not found)
409	Conflict	Duplicate resource (username or email already registered)
422	Unprocessable Entity	Request body fails Pydantic validation (missing fields, wrong types)
429	Too Many Requests	Rate limit exceeded on /auth/login (10 requests/minute per IP)
500	Internal Server Error	Unexpected server error

Swagger UI: <https://t20-datahub-api.onrender.com/docs> · Frontend: <https://t20-datahub-api.onrender.com>